

Supply Chain Performance Analysis: A Data-Driven Review

Final Project Findings & Strategic Recommendations

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Executive Summary: Three Core Findings from Our Analysis

A comprehensive analysis of our supply chain data has revealed critical insights into performance drivers, operational challenges, and strategic opportunities. The objective was to support data-driven decisions to enhance sales, inventory, and shipping.



Skincare Dominates the Portfolio

The skincare category is the primary revenue engine, accounting for nearly half of total revenue (£1.1M), indicating its strategic importance.



Significant Stock Violations Impact Availability

We recorded 98 distinct stock violations, highlighting a critical challenge in inventory management that directly affects product availability and potentially sales.



Supplier Performance is Highly Variable

A significant trade-off exists between supplier lead times and quality. Our most-used supplier also has one of the highest defect rates, presenting an opportunity for optimisation.

Our Four-Week Analytical Process Ensured a Rigorous, Structured Approach

01



Week 1 - Data Foundation: Data Modelling & Preparation

Built a robust data model, then cleaned and preprocessed the dataset to ensure accuracy and consistency. The final dataset was complete with no missing values.

02



Week 2 - Analysis Phase: Answering Critical Business Questions

Formulated and answered key questions about revenue drivers, supplier performance, and operational efficiency using SQL and Python.

03



Week 3 - Forecasting Phase: Predicting Future Trends

Developed forecasting models to project future revenue revenue and stock levels, identifying emerging trends trends based on historical data.

04



Week 4 - Visualisation & Reporting: Delivering Actionable Insights

Consolidated all findings into an interactive dashboard and and this final presentation to facilitate strategic decision-making.

The Data Cleaning Journey: Preparing Supply Chain Data for Analysis

Understanding the Dataset



100 Records, 28 Columns
A concise dataset containing both numeric and categorical information for analysis

A Complete Supply Chain Overview

Contains data on products, suppliers, manufacturing, and logistics to analyse overall efficiency



Initial State: Clean and Complete

The raw data was found to have no missing values or duplicate rows.

The Preprocessing Pathway

Step 1: Standardise & Validate
Formats for names, numbers, and dates were unified and data types were validated

Date
YYYY-MM-DD
DD/MM/YYYY
DD/MM/YYYY

Date
YYYY-MM-DD
DD/MM/YYYY
...

NAME

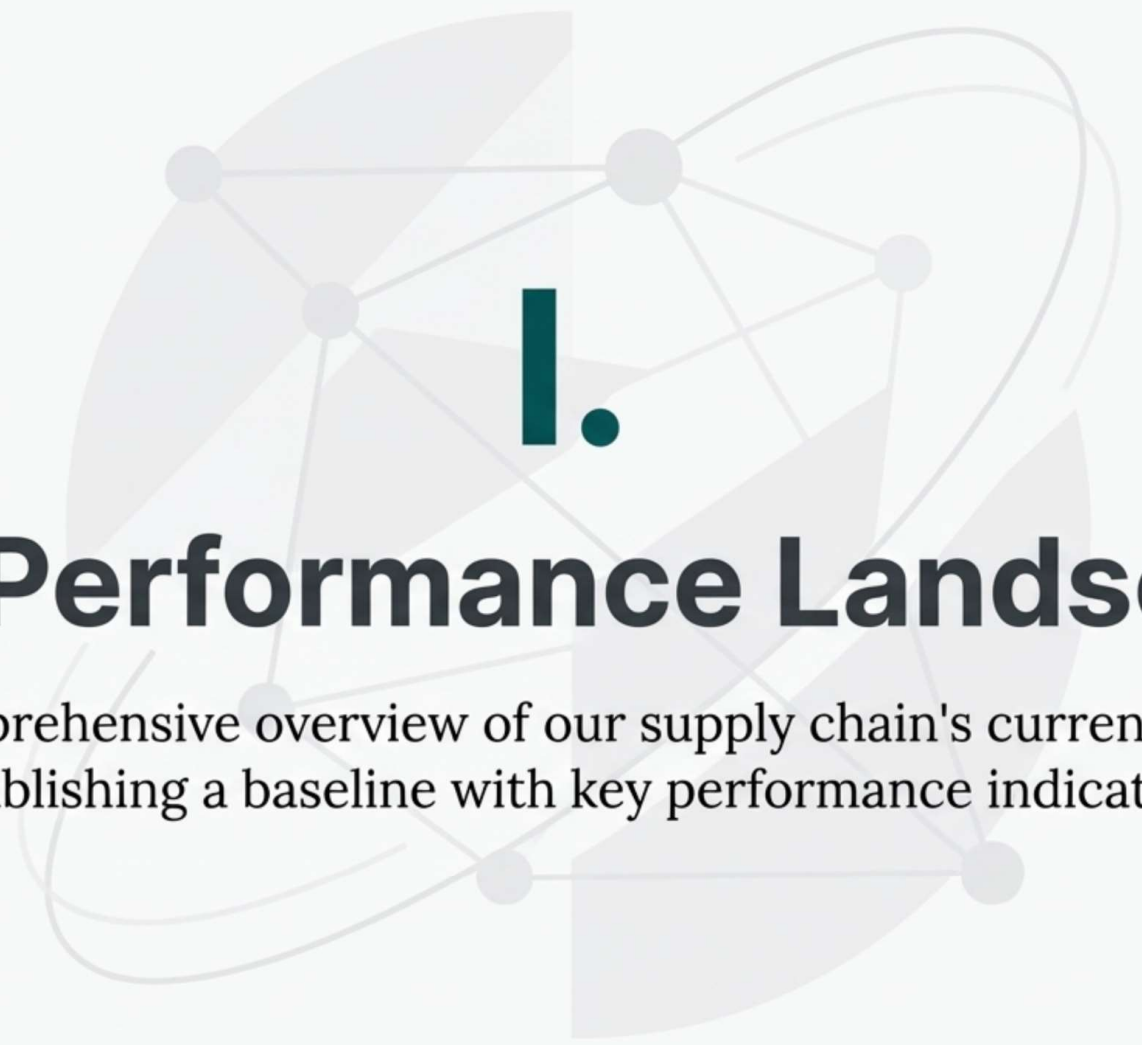


Step 2: Create Calculated Fields
New fields like "Revenue Difference" were created to enable deeper analysis



Result: An Analysis-Ready Asset
The final dataset is fully clean, structured, and suitable for forecasting.





I.

The Performance Landscape

A comprehensive overview of our supply chain's current state, establishing a baseline with key performance indicators.

At a Glance: Key Performance Indicators Show Solid Revenue but Highlight Operational Drag



£2.29M

Total Revenue Generated



46K

Total Products Sold



17 Days

Average Supplier Lead Time

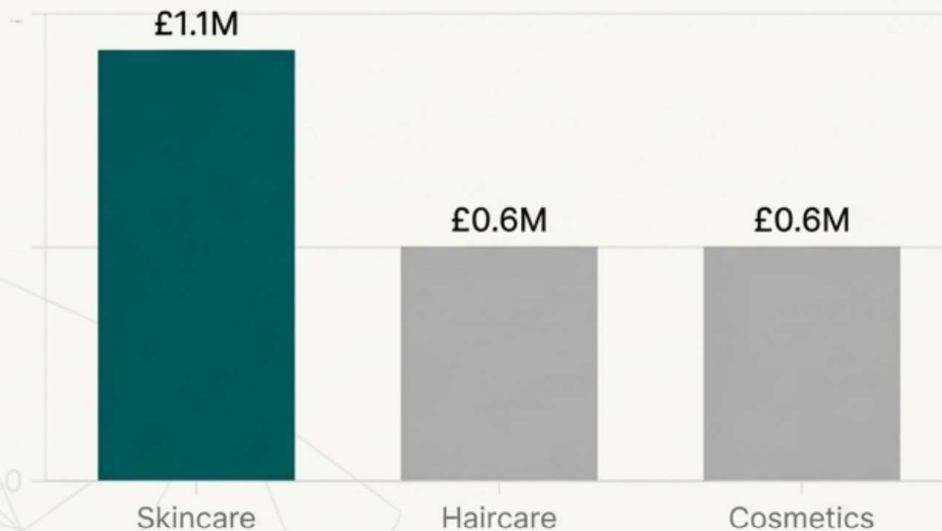


98 Incidents

Recorded Stock Violations

Skincare Products are the Clear Revenue Engine, Contributing £1.1M to the Top Line

Revenue by Product Category



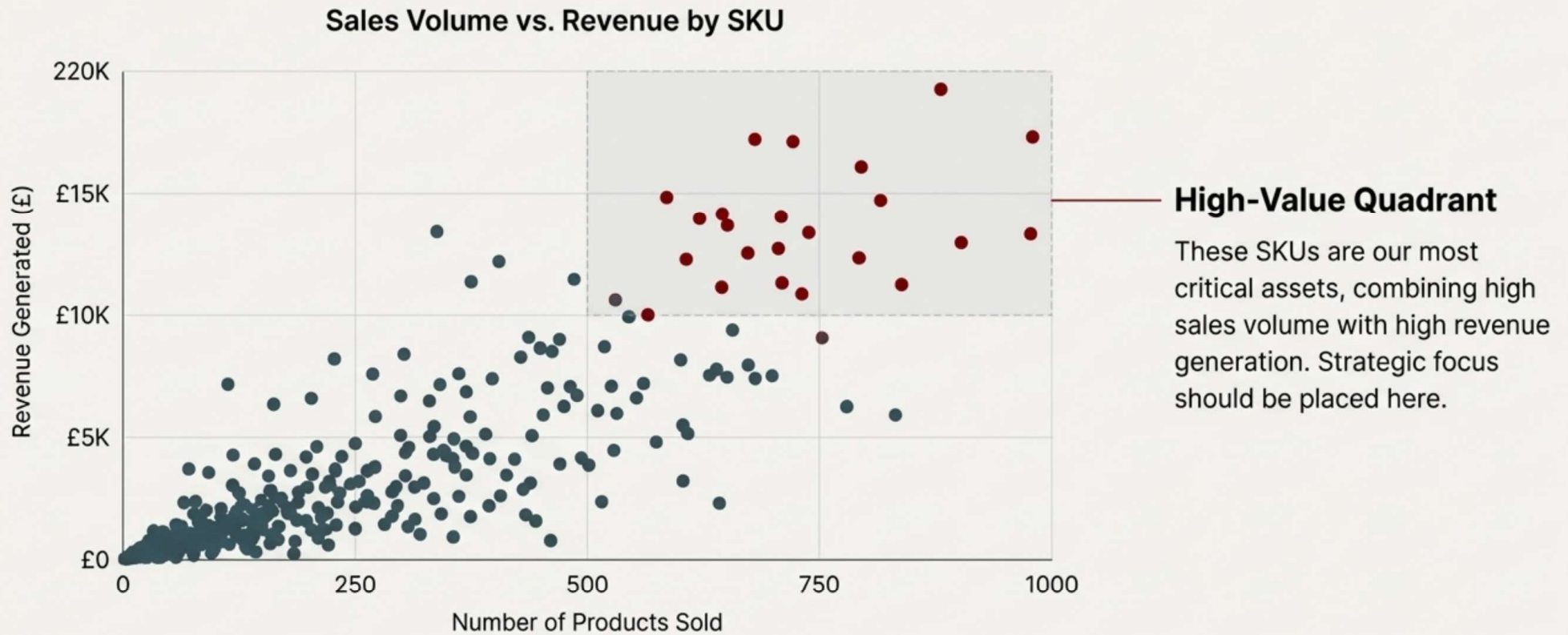
Category Contribution

While making up 40% of the product types in the dataset, the skincare category generates nearly half of the total revenue.

Haircare (34% of products) and Cosmetics (26%) contribute almost equally to the remaining revenue.

This concentration of revenue highlights the critical importance of **maintaining supply chain health and product availability** for the skincare line.

Identifying Our Star Performers: Mapping Sales Volume to Revenue by SKU



Our data allows us to pinpoint not just popular products, but profitable ones. The next step is to ensure these high-performing SKUs are always available for our customers.

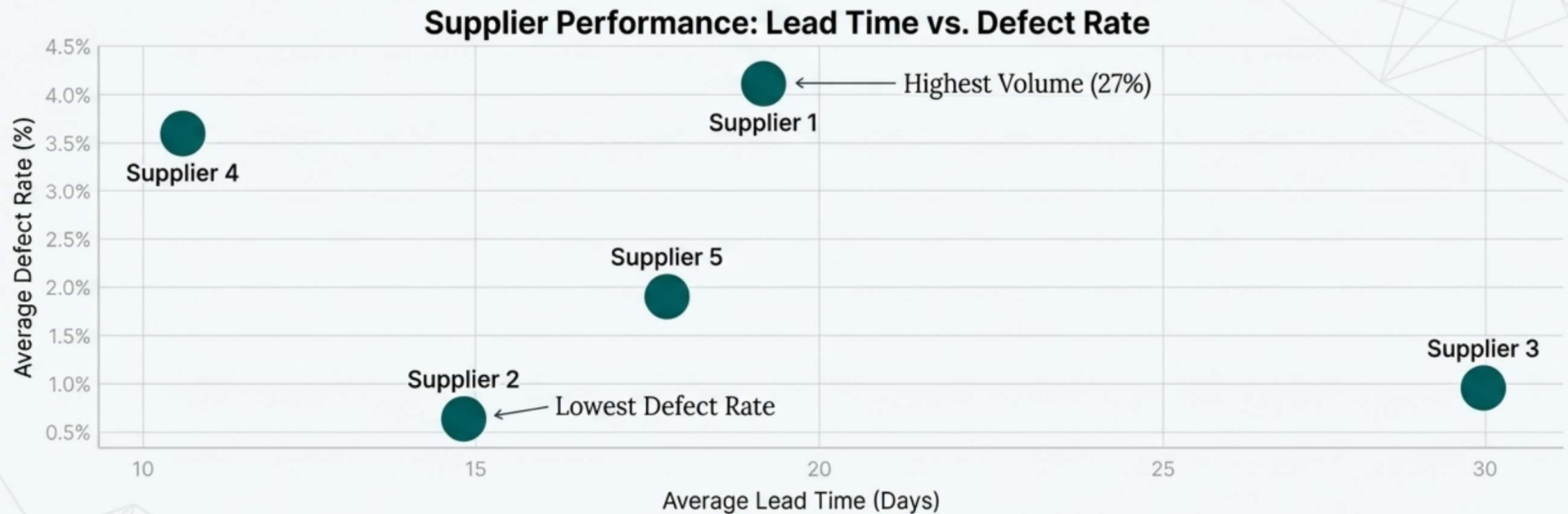
A faint, light blue Venn diagram with three overlapping circles is centered in the background of the slide.

II.

Partner Ecosystem Analysis

Evaluating the performance and efficiency of our key suppliers and shipping carriers.

Supplier Performance Reveals a Critical Trade-Off Between Defect Rate and Lead Time



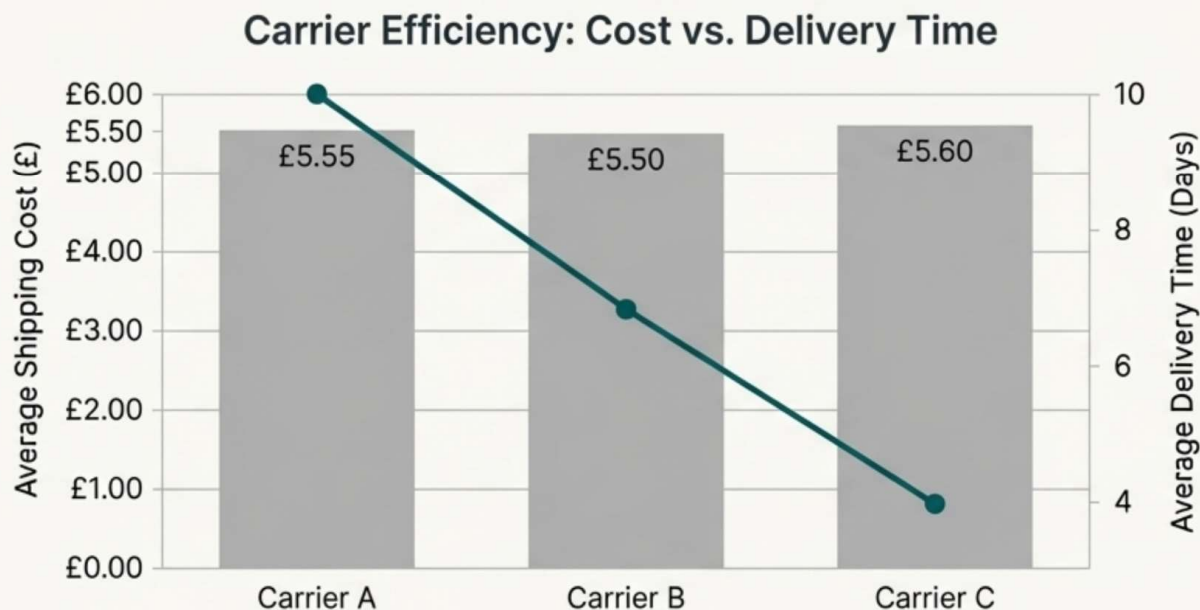
Our Highest Volume Partner Presents a Quality Risk

Supplier 1 is our most frequent partner, used in 27% of cases. However, they exhibit one of the highest average defect rates, creating downstream quality control costs and risks.

Opportunity for Optimisation

Supplier 2 demonstrates the lowest defect rate in the network. Exploring opportunities to shift volume to higher-quality partners like Supplier 2 could significantly improve overall product quality.

Carrier Efficiency Analysis Shows Delivery Speed is the Key Differentiator, Not Cost



Performance Profile

While shipping costs are nearly identical across all carriers, there is a distinct performance gap in delivery times. Carrier B is our most used partner (43% of shipments), offering a moderate delivery speed.

Strategic Implication

This presents an opportunity to optimise our shipping strategy by allocating shipments based on urgency vs. cost, potentially using the faster Carrier C for priority orders.

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III.

Operational Health Check

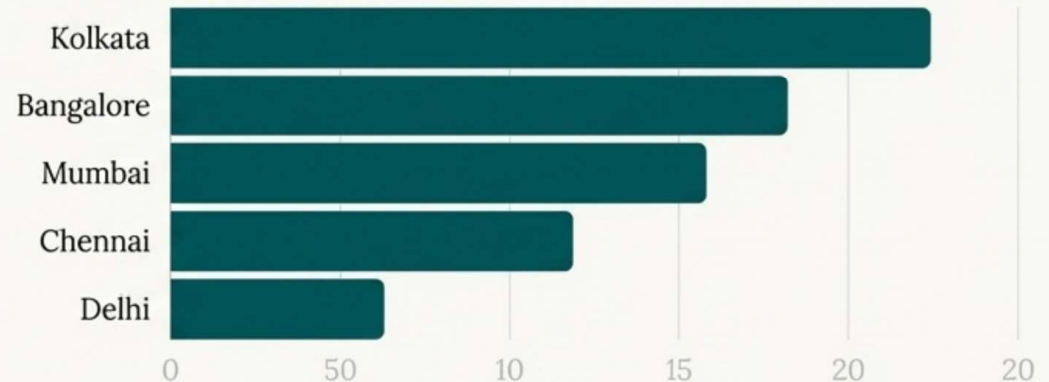
Assessing the effectiveness of our internal inventory management and identifying key operational challenges.

Inventory Management is a Key Challenge, with 98 Stock Violations Recorded

98

Total Stock Violations

Stock Levels by Location



The Impact of Violations

These 98 violations represent instances where stock levels fell below required thresholds, leading to potential stock-outs, lost sales, and reduced customer satisfaction. Certain SKUs are repeatedly affected.

The disparity in stock levels across locations, with Delhi holding the least inventory, suggests a need to review our inventory allocation strategy to mitigate violation risks.

The background of the slide features a light gray, abstract design. It includes a series of concentric, slightly irregular circles that create a sense of depth and movement, resembling a road or a path. Overlaid on these are several thin, dark gray lines that intersect at various angles, forming a geometric pattern that suggests a network or a complex system.

IV.

The Road Ahead

Leveraging historical data to forecast future performance and formulate data-driven strategic recommendations.

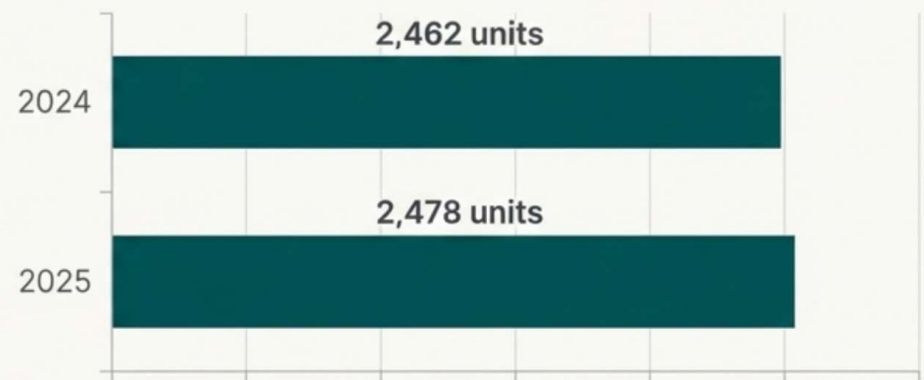
Forecasts Project Stable Stock Levels and Fluctuating Short-Term Revenue

Revenue Forecast for Upcoming Periods



Note: Represents a periodic forecast, not full annual revenue.

Forecasted Stock Levels



The forecast indicates a slight increase in inventory levels, suggesting that stock-outs may not worsen systemically. However, the short-term revenue forecast shows some volatility, reinforcing the need for proactive management of our key product categories.

Strategic Recommendations to Enhance Performance



1. Reinforce the Core Business

Double down on the skincare category. Prioritise its stock allocation, supplier quality, and marketing support to protect and grow our primary revenue stream.



2. Optimise the Supplier Mix

Initiate a supplier performance review. Develop a strategy to shift volume from high-defect partners (e.g., Supplier 1) to high-quality partners (e.g., Supplier 2) to reduce defect-related costs.



3. Address Stock Violations Head-On

Launch a root-cause analysis for the 98 stock violations, focusing on the most frequently affected SKUs and lowest-stocked locations like Delhi. Re-evaluate safety stock levels.



4. Refine Logistics Strategy

Implement a tiered shipping strategy. Use the most cost-effective carriers (e.g., Carrier B) for standard shipments and reserve faster options (e.g., Carrier C) for high-priority or time-sensitive orders.

Next Steps: Embedding Data-Driven Decision-Making



Schedule Stakeholder Workshops

We propose a series of deep-dive workshops with relevant teams (Supply Chain, Sales, Operations) to discuss these findings and collaboratively develop detailed action plans for each recommendation.



Leverage the Live Dashboard

The accompanying Tableau dashboard provides a tool for ongoing, real-time monitoring of these KPIs. We will provide training to ensure it is integrated into regular performance reviews.

This analysis provides a clear roadmap for optimising our supply chain. We look forward to partnering with you to implement these changes.

Supply Chain Performance: A Data-Driven Overview



Overall Performance Snapshot

**2.29M in
Total Revenue**

Generated from the sale of over
46,000 individual products.



**98 Stock Violations
Recorded**

Highlighting a critical challenge in
current inventory management.

Operational Insights & Future Outlook



**Supplier 2 Has
the Lowest
Defect Rates**

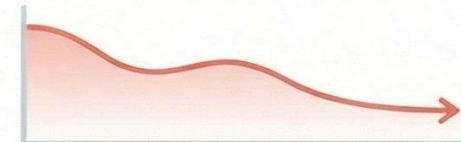
Making them the most
reliable partner for ensuring
product quality.



**Carrier B is the
Most Used and
Cost-Effective
Shipper**

Handling 43% of
shipments while
maintaining competitive
shipping costs.

**Revenue is Forecast
to Dip in 2025**



Projections show a slight decrease from
2024 levels, requiring strategic focus.

Revenue by Category

Skincare:
1.1M

Haircare:
0.6M

Cosmetics:
0.6M